Experiment - 2

Learning and assignment of IP address Manually to Computers

Aim:

To Understand the IP addressing scheme and manually to Computers on a network

procedure:

1. In Windows: Open Network and Sharing Center → Change Adapter Settings → Right-Click Local Area Connection → Properties → IPv4 → Set IP manually

2. Enter IP Address: 192.168.1.10, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1

3. Enter DNS Server: 8.8.8.8, click Ok

4. In Linux: Use command 'sudo ip addr add 192.168.1.10/24 dev eth0'.

5. To set gateway in Linux: 'sudo ip route add default via 192.168.1.1'

6. For persistent configuration in Linux, edit /etc/network/interfaces or use nmcli

7. Ping the default gateway to confirm

8. Ping another computer on the same subnet to verify communication

Code :

```
Sudo ip addr add 192.168.1.10/24 dev eth0
sudo ip route add default via 192.168.1.1
ip a show eth0

ping 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.addresses 192.168.1.10
/24
nmcli con mod 'Wired connection 1' ipv4.gateway 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.method manual
nmcli con up 'Wired connection 1'
```

Result

Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.